



Spatial Targeting of Small-scale Small Ruminant Silvopastoral Systems in Ghana

Where near Tamale could smallholder small-ruminant silvopastoral systems be piloted?

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Introduction



- ▶ Silvopastoralism is a type of agroforestry that combines livestock grazing with trees within the same production unit, leveraging the ecological relationships between animals and woody vegetation. These systems enhance land productivity, support biodiversity, increase carbon sequestration and bring long term social and economic advantages to the farmer. (FAO, 2022; Montagnini et al., 2013)
- ▶ In Ghana, climate change has manifested through declining rainfall, rising temperatures, rising sea levels and increased frequency of extreme weather events. (Adonadaga et al., 2022)
- ▶ This exercise aims to help researchers decide where in Ghana they can feasibly pilot and demonstrate small-ruminant silvopastoral systems. It applies the interactive tools: [The Land Targeting Tools](#).

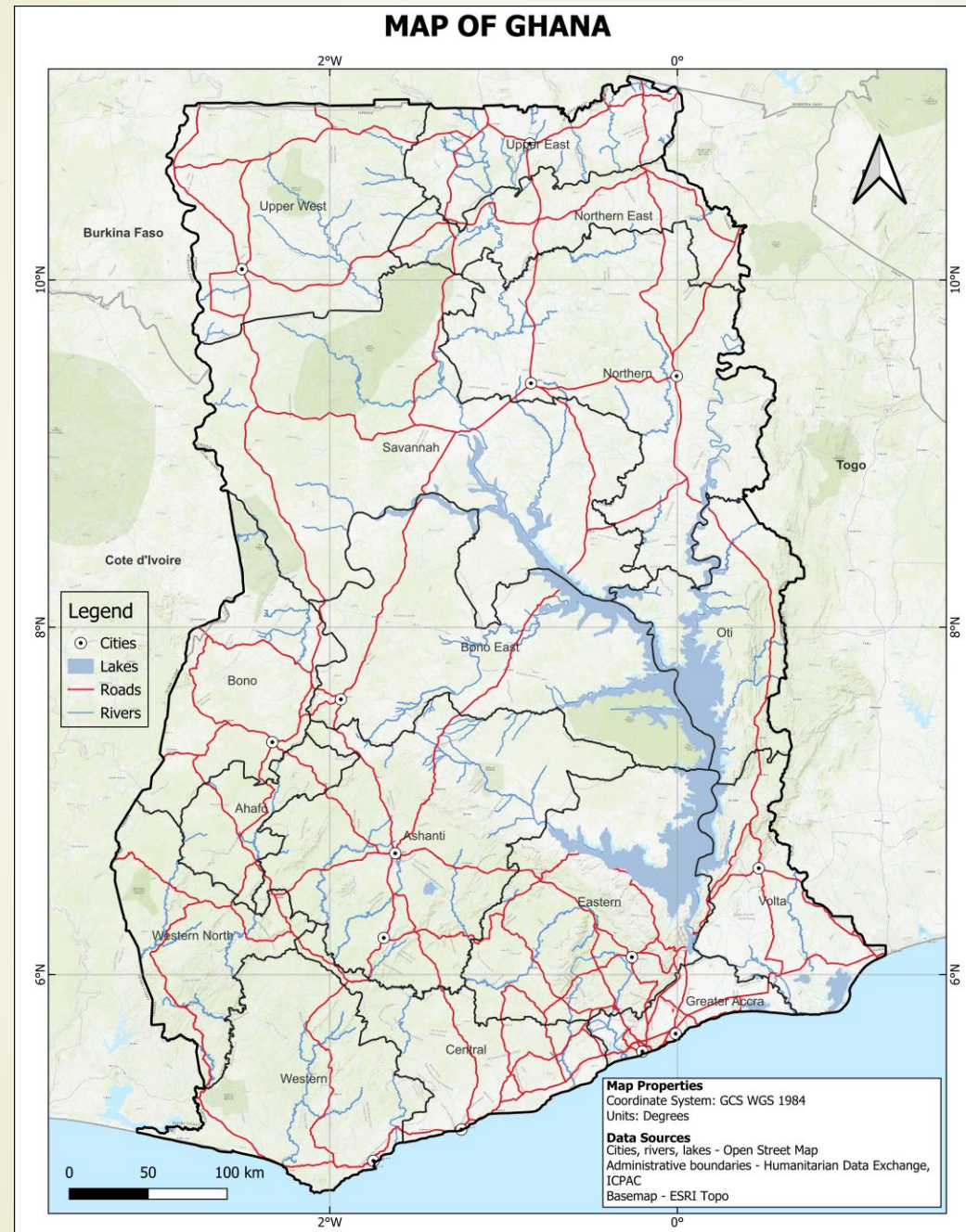


Problem statement



- ▶ Existing hotspot maps mainly focus on biophysical criteria such as climate.
- ▶ Non-biophysical factors that affect implementation, such as accessibility and labour, are often excluded, limiting the practicability of testing, piloting and scaling interventions.
- ▶ Scientists piloting silvopastoral systems in Ghana require accurate spatial guidance that goes beyond the suitability of biophysical factors.

Study Area



- The Republic of Ghana is located in West Africa, neighbours being Cote d'Ivoire to the west, Burkina Faso to the north and Togo to the east.
- Ghana is characterized by 6 distinct agro-ecological zones (guinea savannah, sudan savannah, transition zone, rain forest, deciduous forest, and coastal savannah).
- Ghana has high annual rainfall varying from 800 to 2,200 mm.

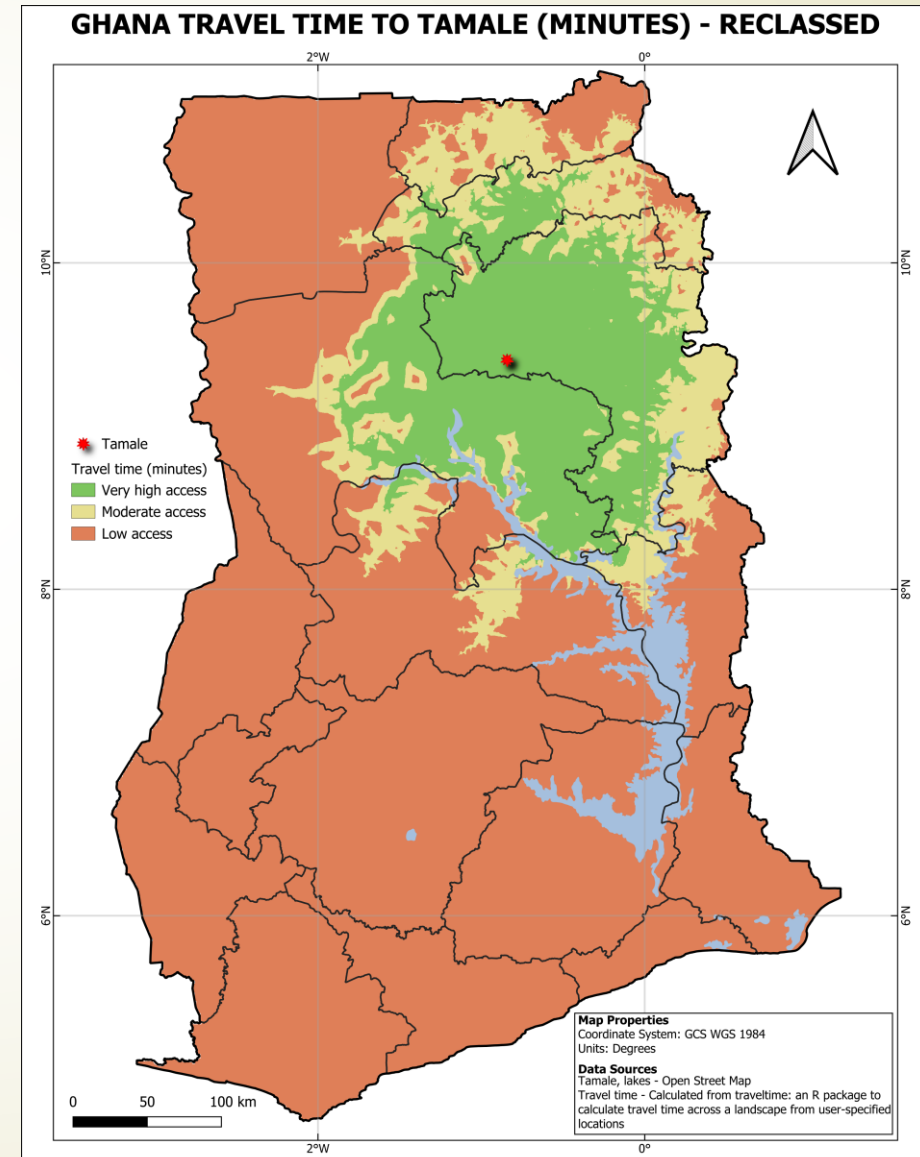
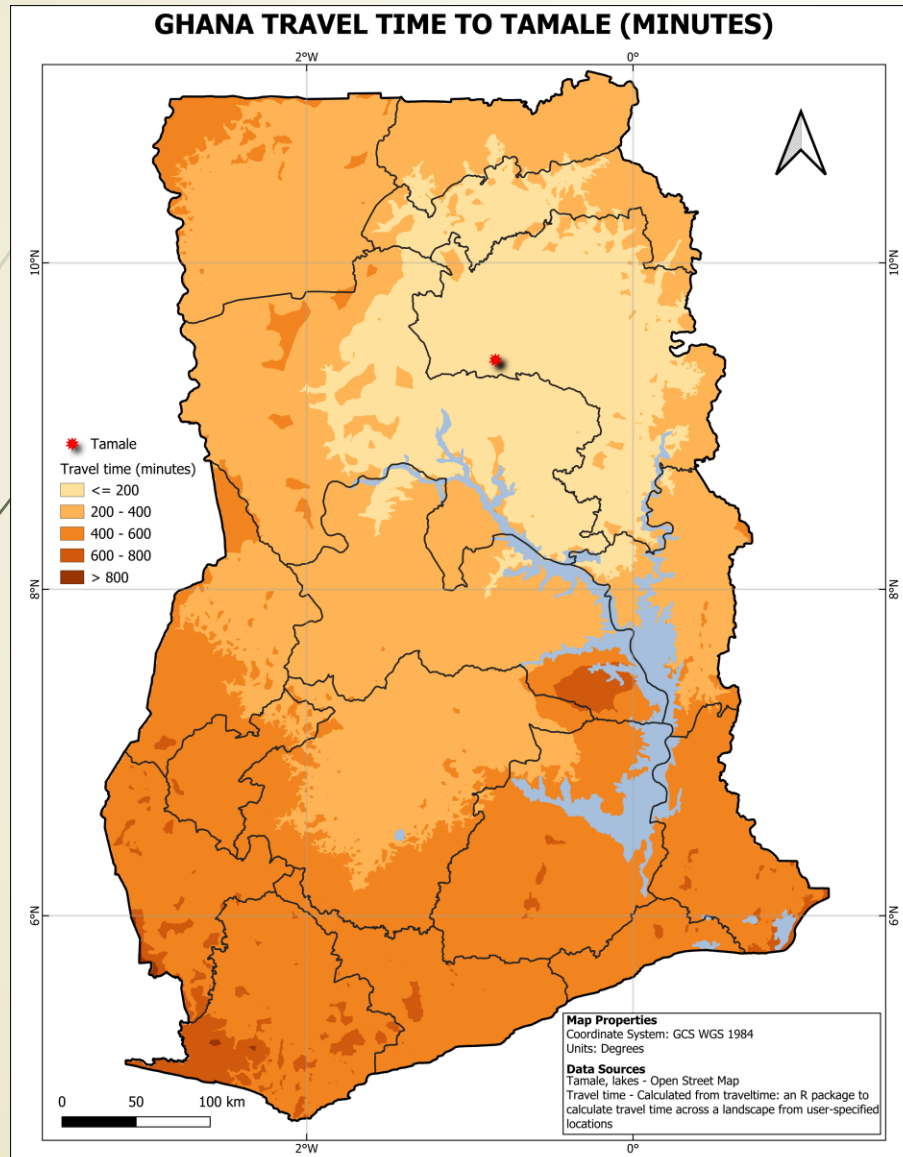
Datasets

Data

Dataset	Source
Accessibility to Tamale (minutes)	Calculated using R Package: traveltime
Family labour – Female above 15 years population density	Worldpop - Age sex grid data
Small ruminant livestock density	Gridded Livestock of the World – sheep and goats
Accessibility to urban centres with population >=50k	Calculated using R Package: traveltime
Shade tolerant forage suitability maps (6 species)	Gaisberger et al. 2026
Appropriate tree species suitability maps (2 species)	Tree species suitability datasets, Ghana
Degradation layer – Land Productivity Dynamics	Land Productivity layer (2013-2022)

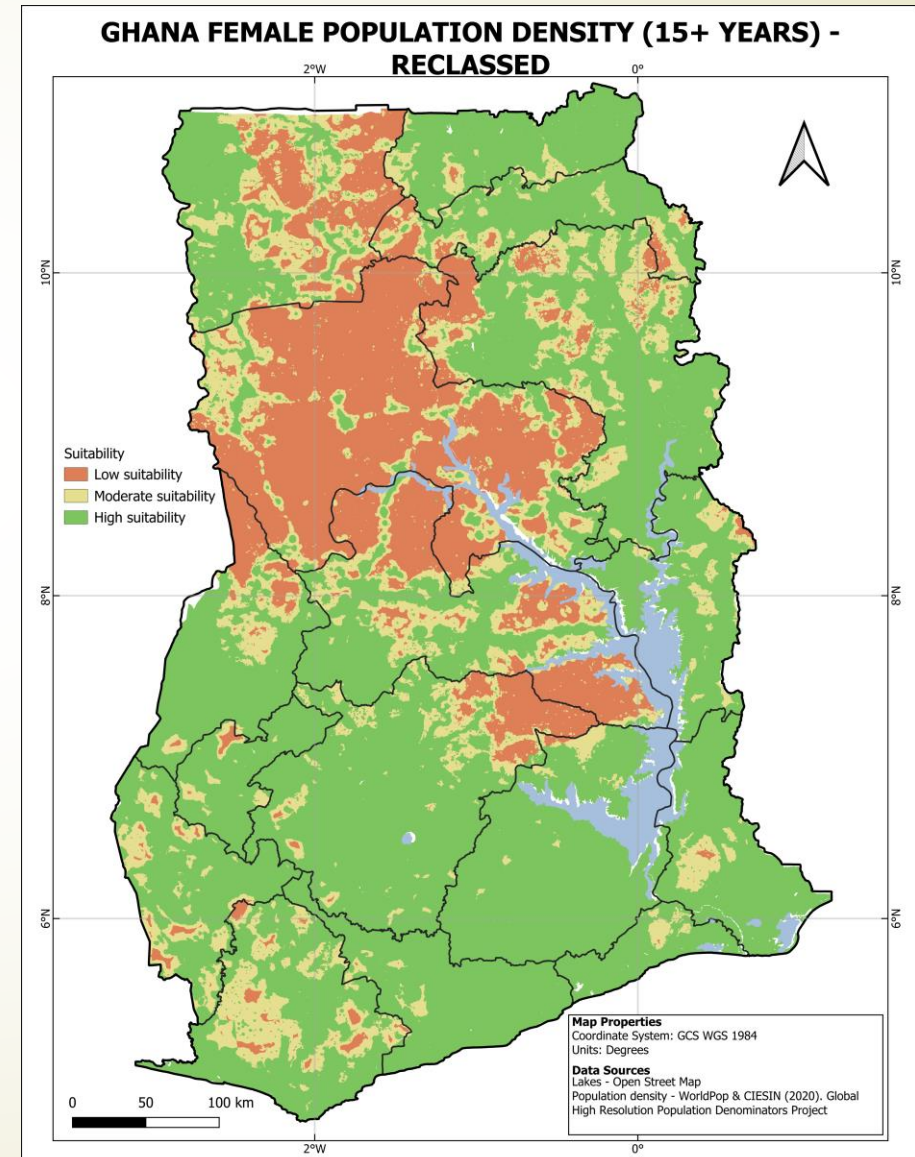
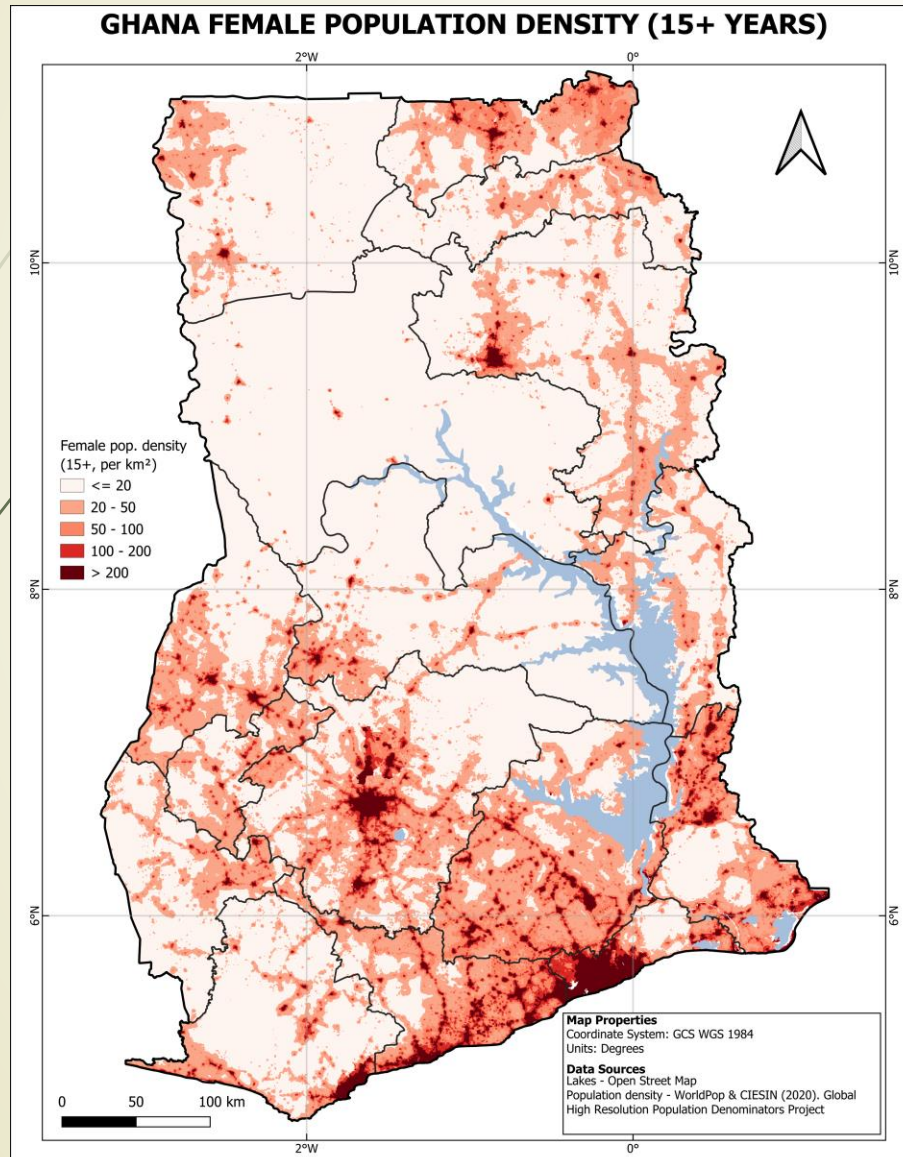
Accessibility

Accessibility is represented by travel-time to Tamale by road. The purpose is to identify areas that can be reached within practical timeframes, helping scientists plan field visits and pilot activities around the region. The optimal class (very high access - green) is set to range from 0 – 180 minutes)



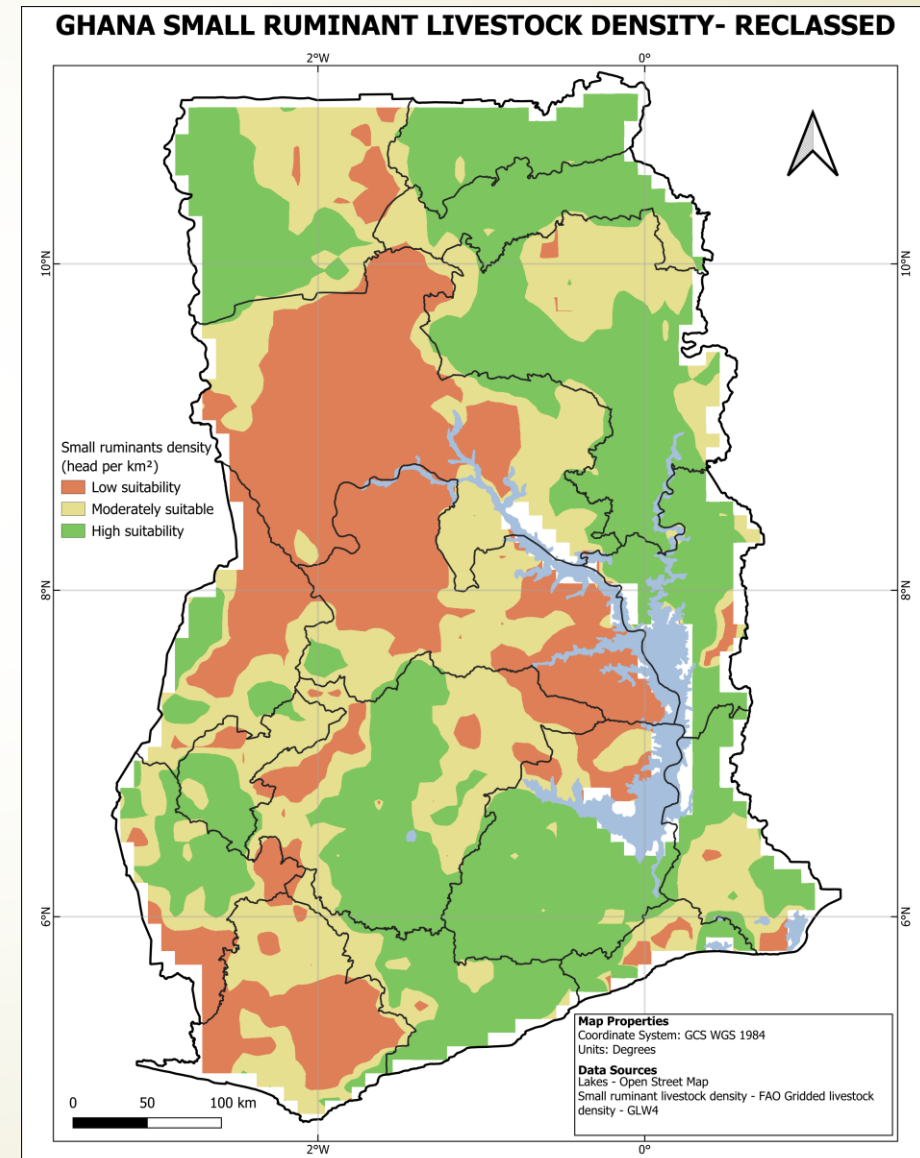
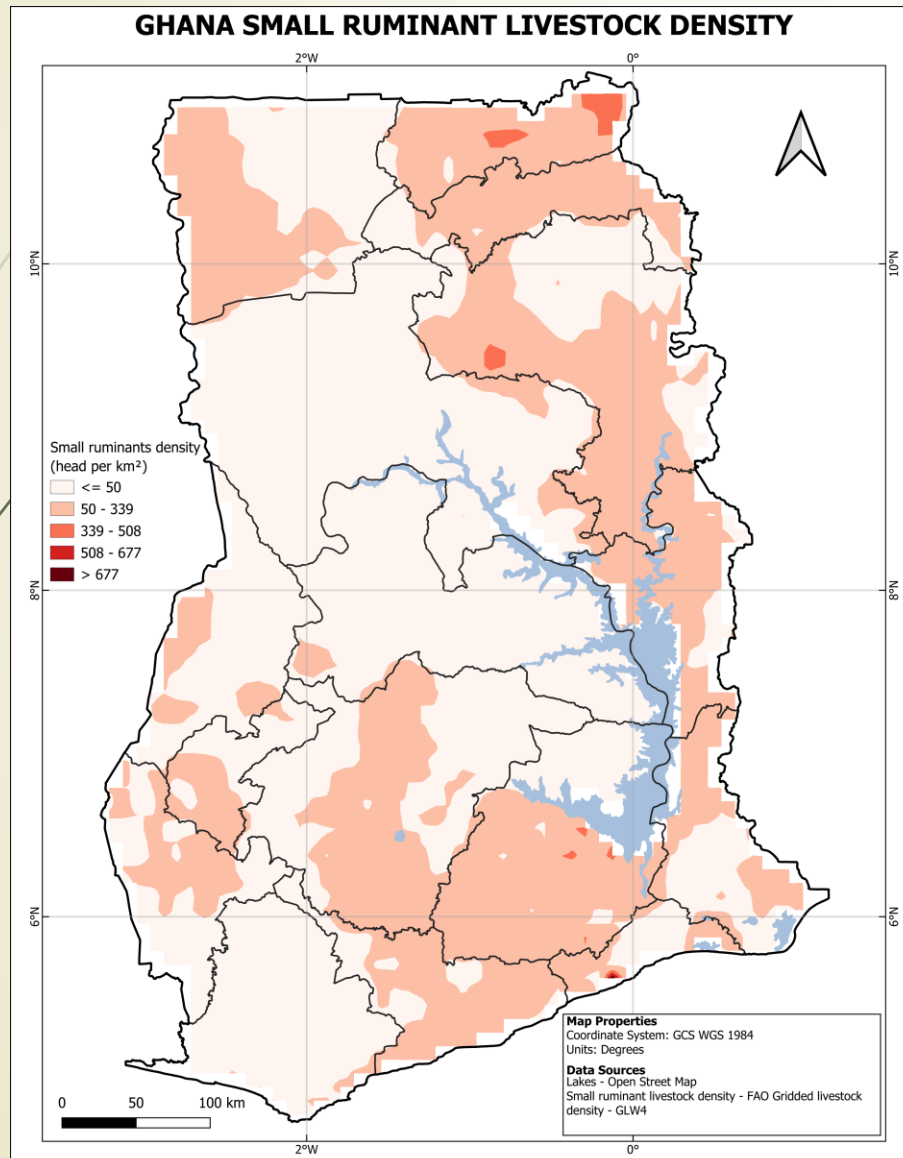
Fam. labour

This represents the density of females over 15 years in Ghana. It serves as a proxy for labour availability since the primary participants in smallholder silvopastoral systems are often women in the region. Optimal class – high suitability (green) - is comprised of areas with more than 10 females/ km²



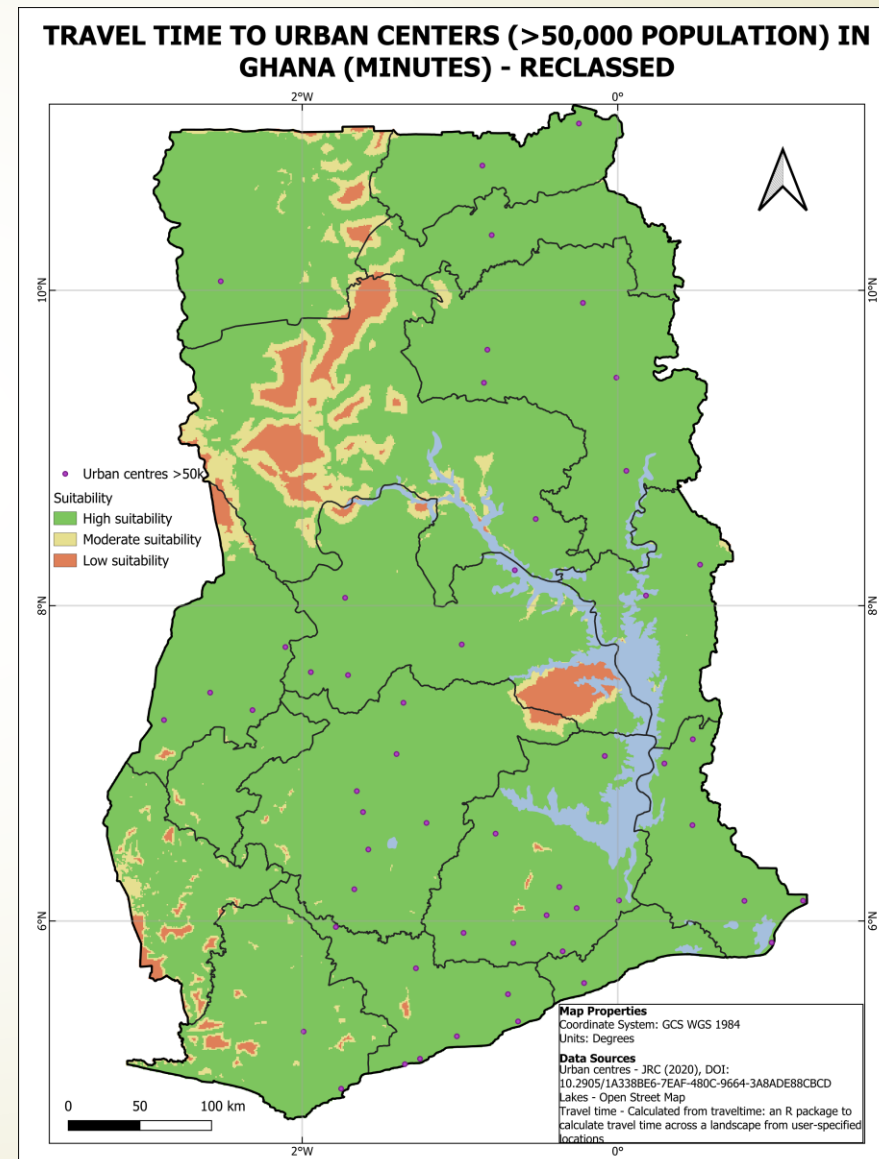
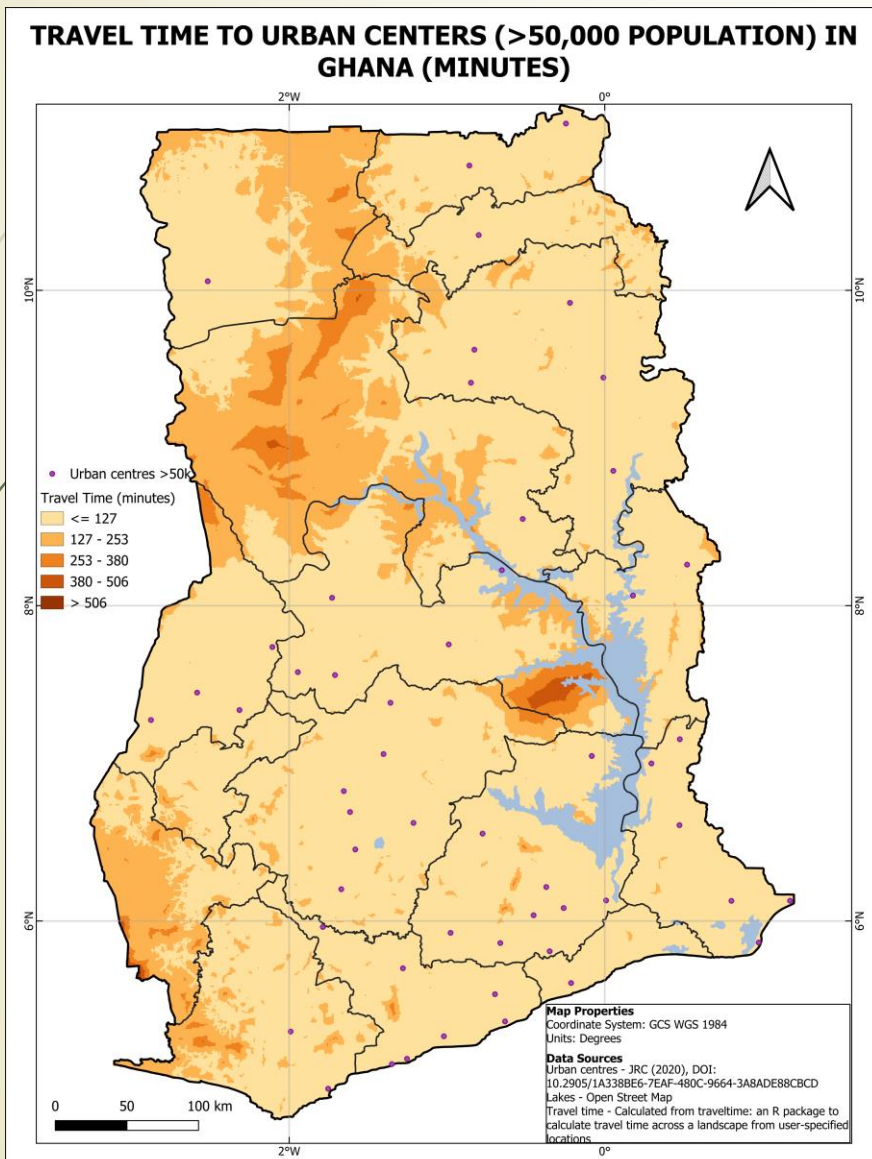
Small ruminants

This represents the density of small ruminants (sheep and goats) in the region. It helps in identifying regions where small-ruminant silvopastoral systems would be of interest. The optimal class (high suitability - green) corresponds to ≥ 50 heads/km²



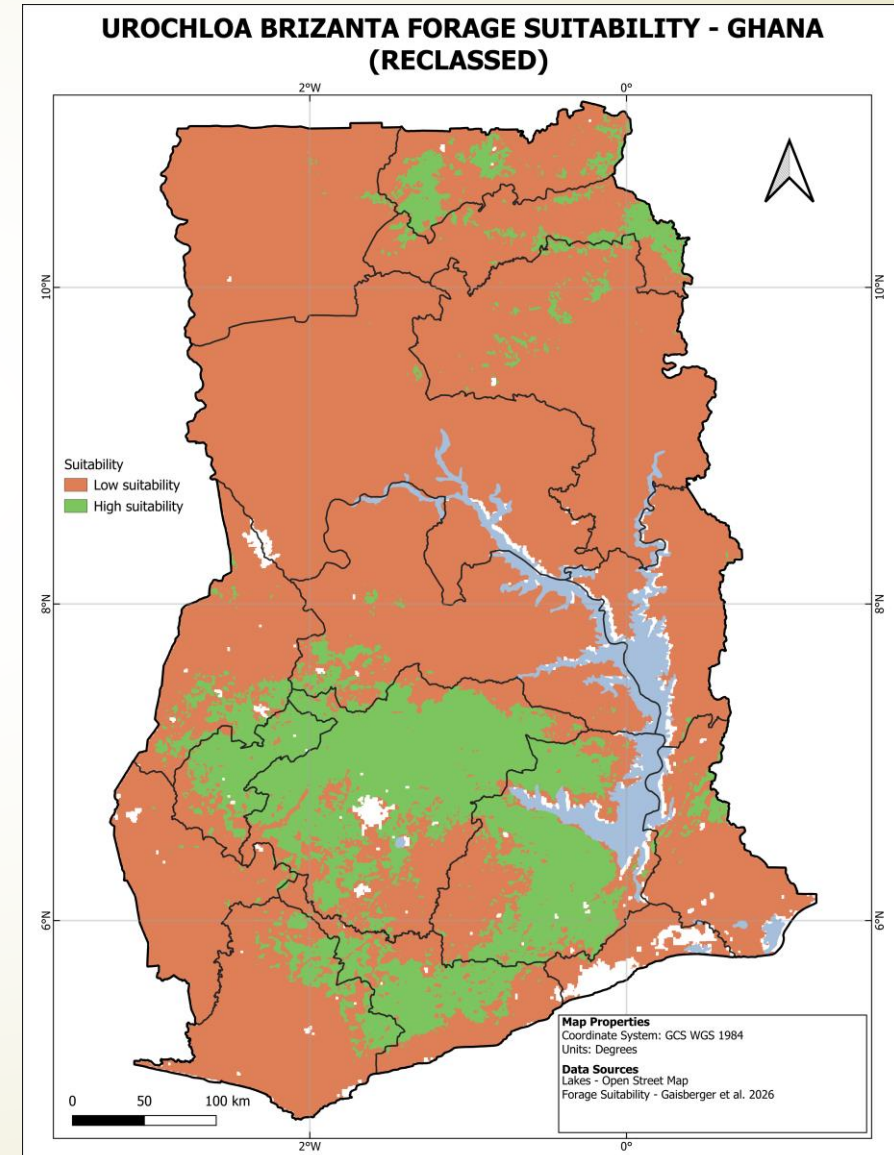
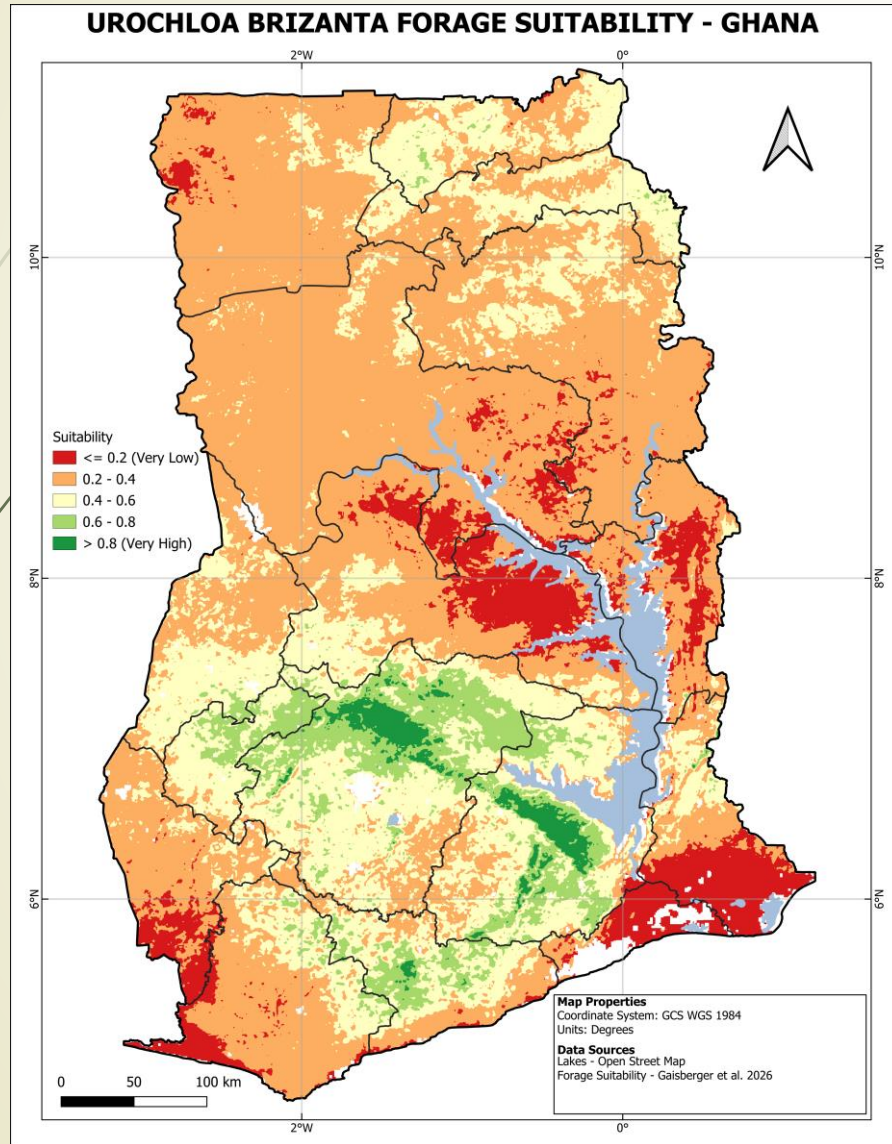
Market access

This is travel time to urban centers (>50k population). It identifies regions with easier market access for the smallholder farmers to sell their products. The optimal classes (very high access - green) ranged from 0 – 180 minutes travel time.



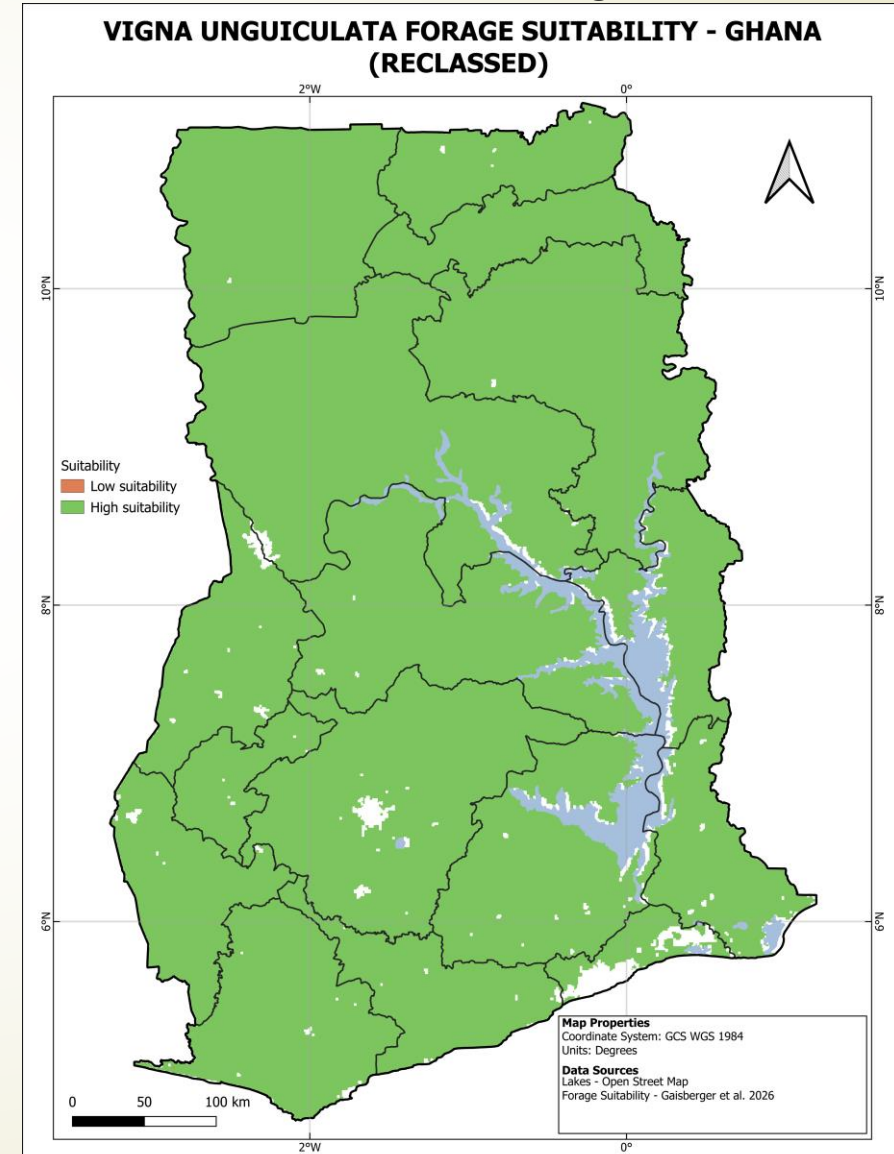
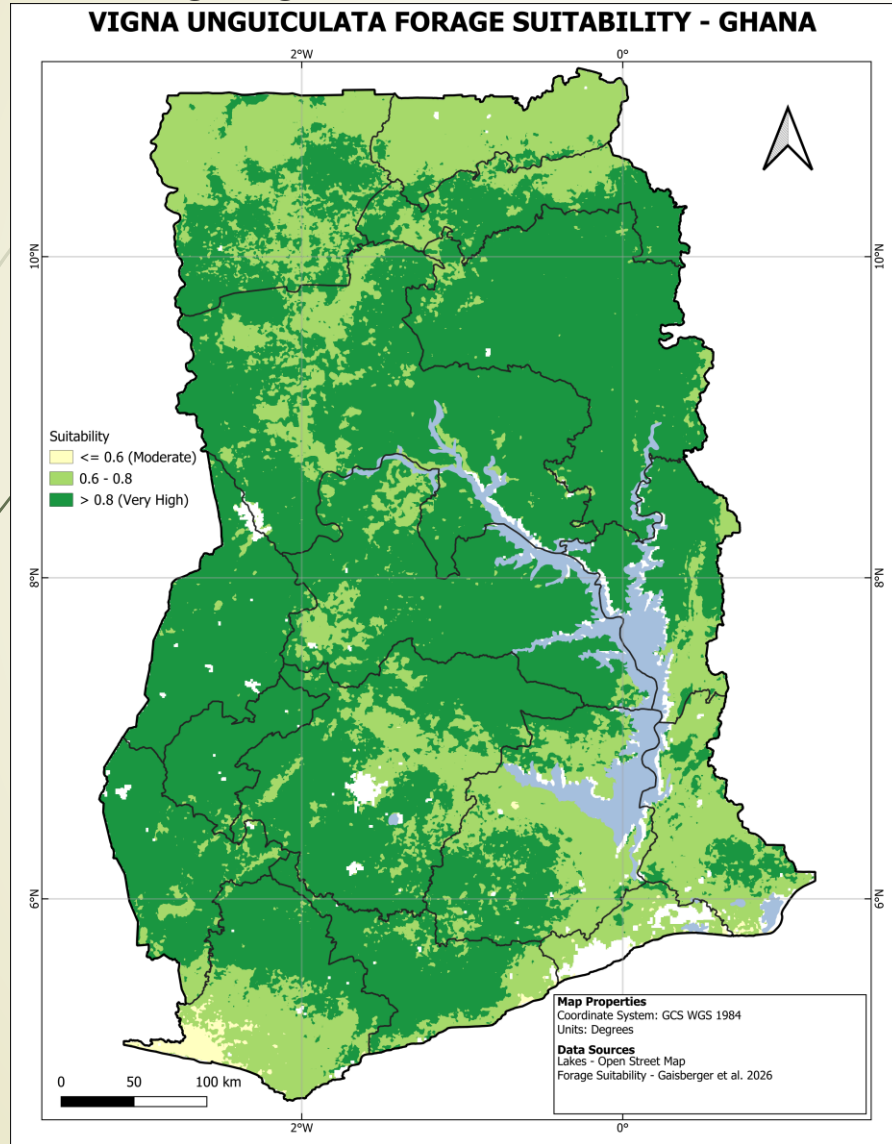
Forages

The forage suitability map help to identify regions suitable for cultivation of shade-tolerant forage species to support livestock production : *Urochloa brizanta*. Areas in the optimal class (green) correspond to a suitability value greater than 0.5, indicating regions most suitable for cultivation of the forage species.



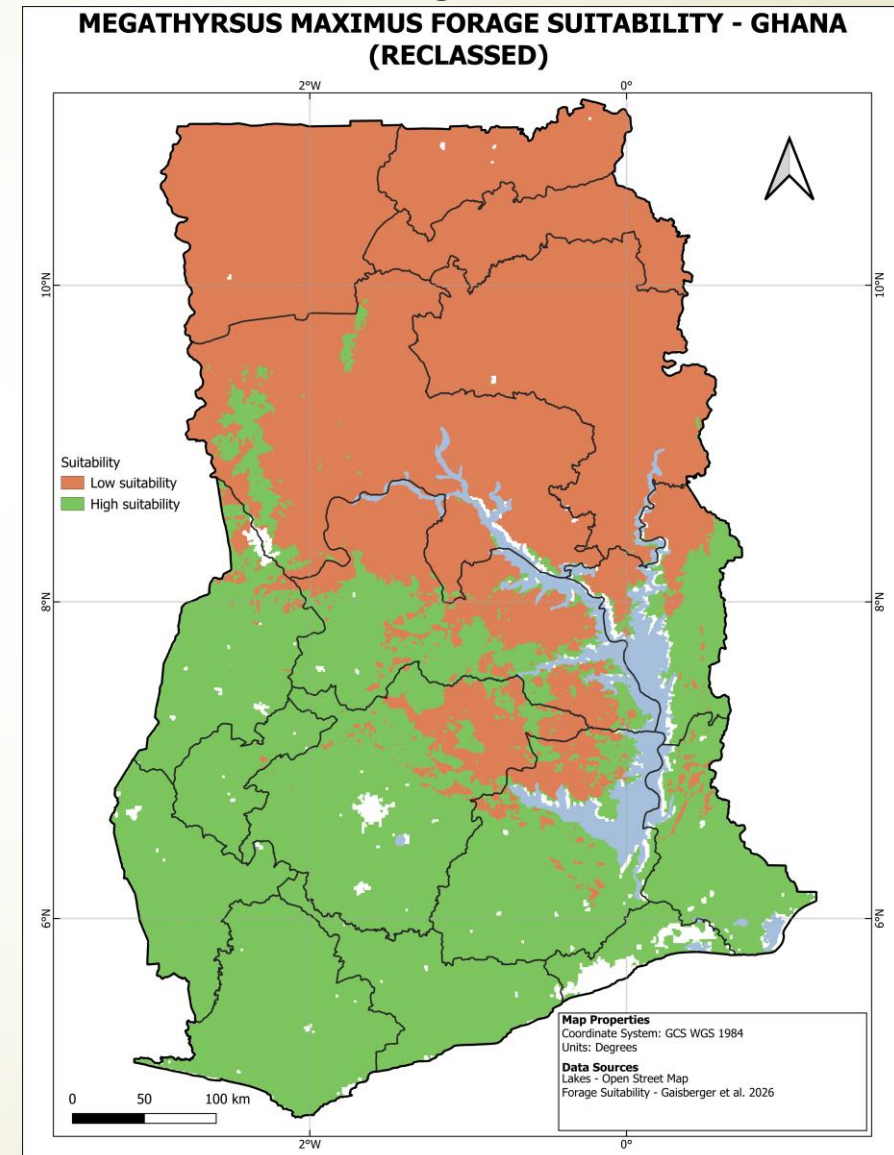
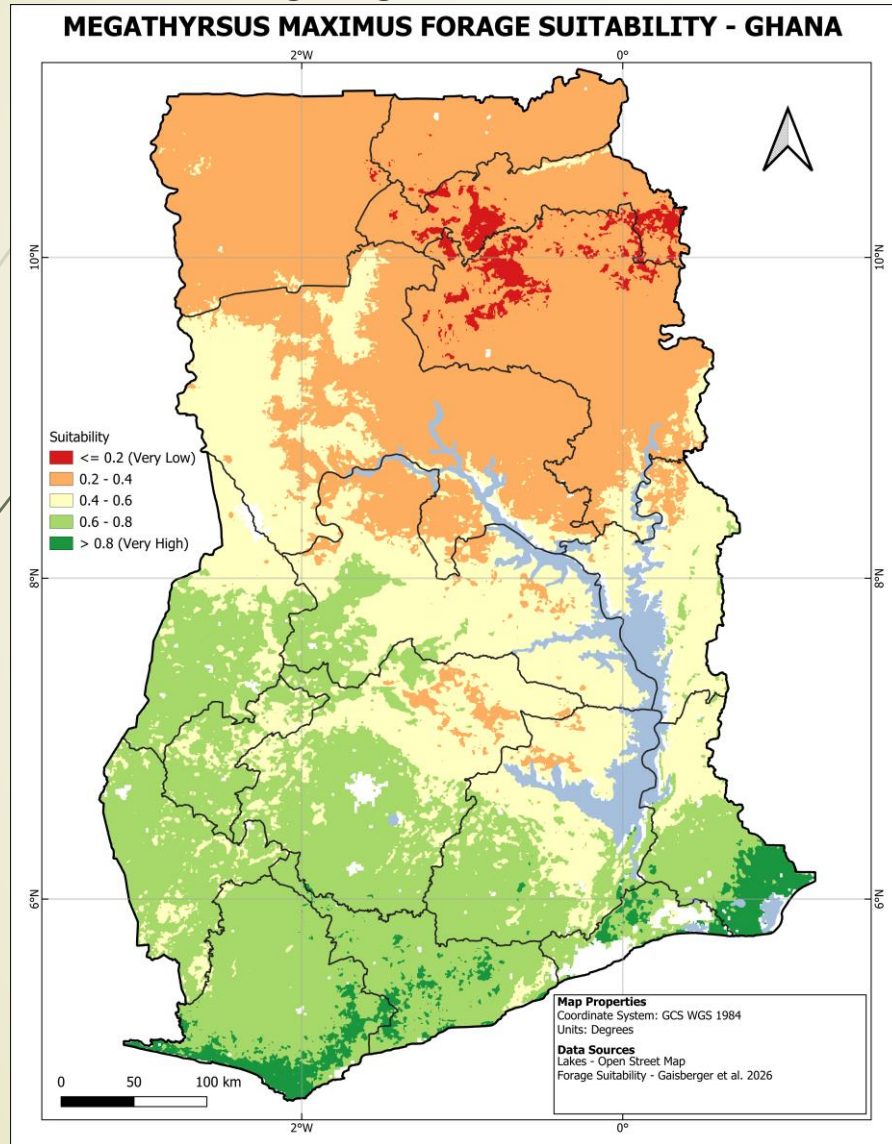
Forages

The forage suitability map help to identify regions suitable for cultivation of a number of shade-tolerant forage species to support livestock production. Here : *Vigna unguiculata*. Areas in the optimal class (green) correspond to a suitability value greater than 0.5, indicating regions that are indeed suitable for cultivation of the forage species.



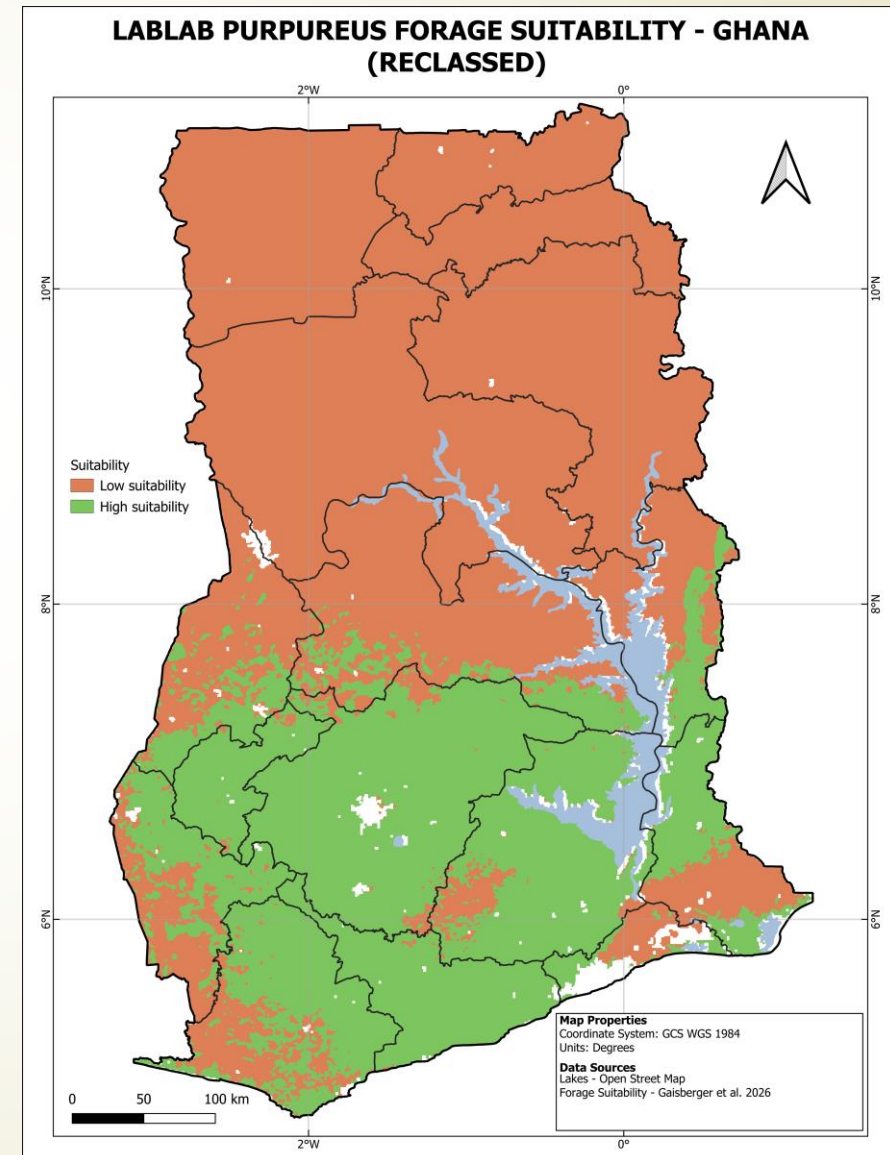
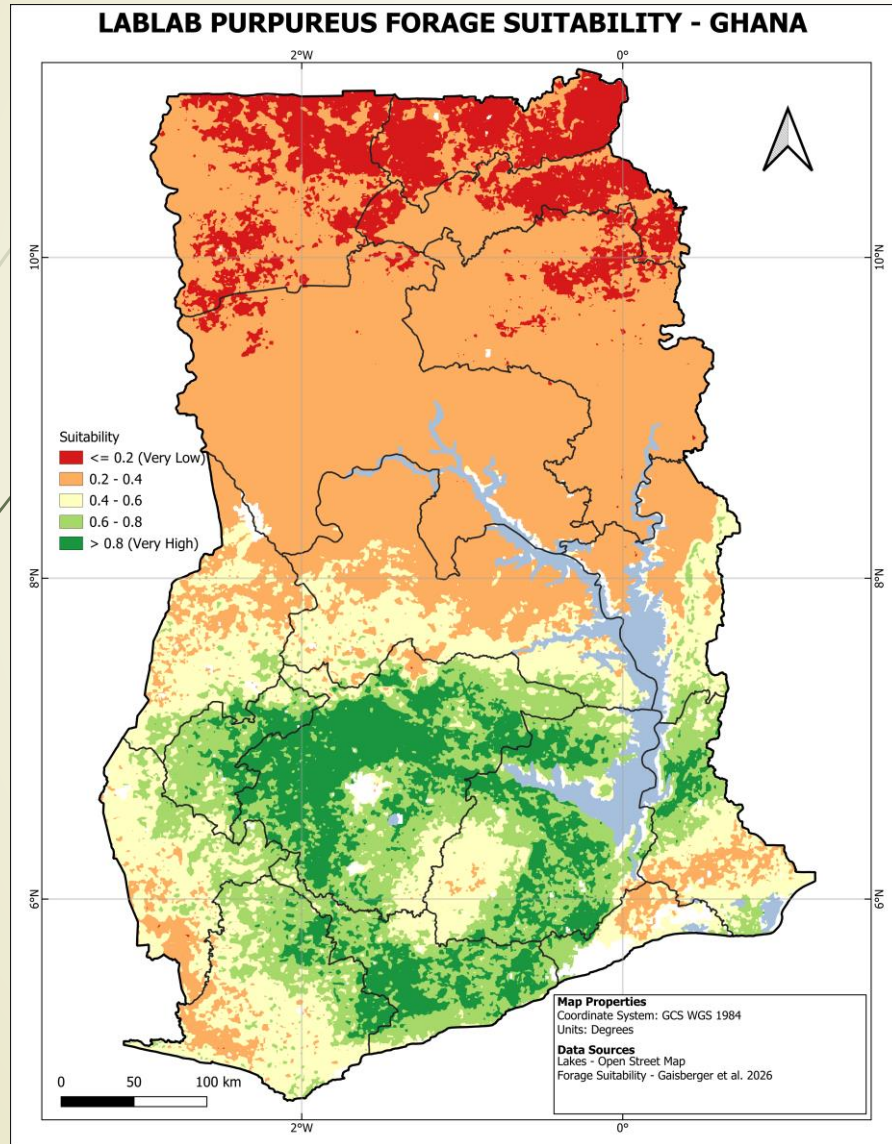
Forages

The forage suitability map help to identify regions suitable for cultivation of a number of shade-tolerant forage species to support livestock production. Here : *Megathyrus maximus*. Areas in the optimal class (green) correspond to a suitability value greater than 0.5, indicating regions most suitable for cultivation of the forage species.



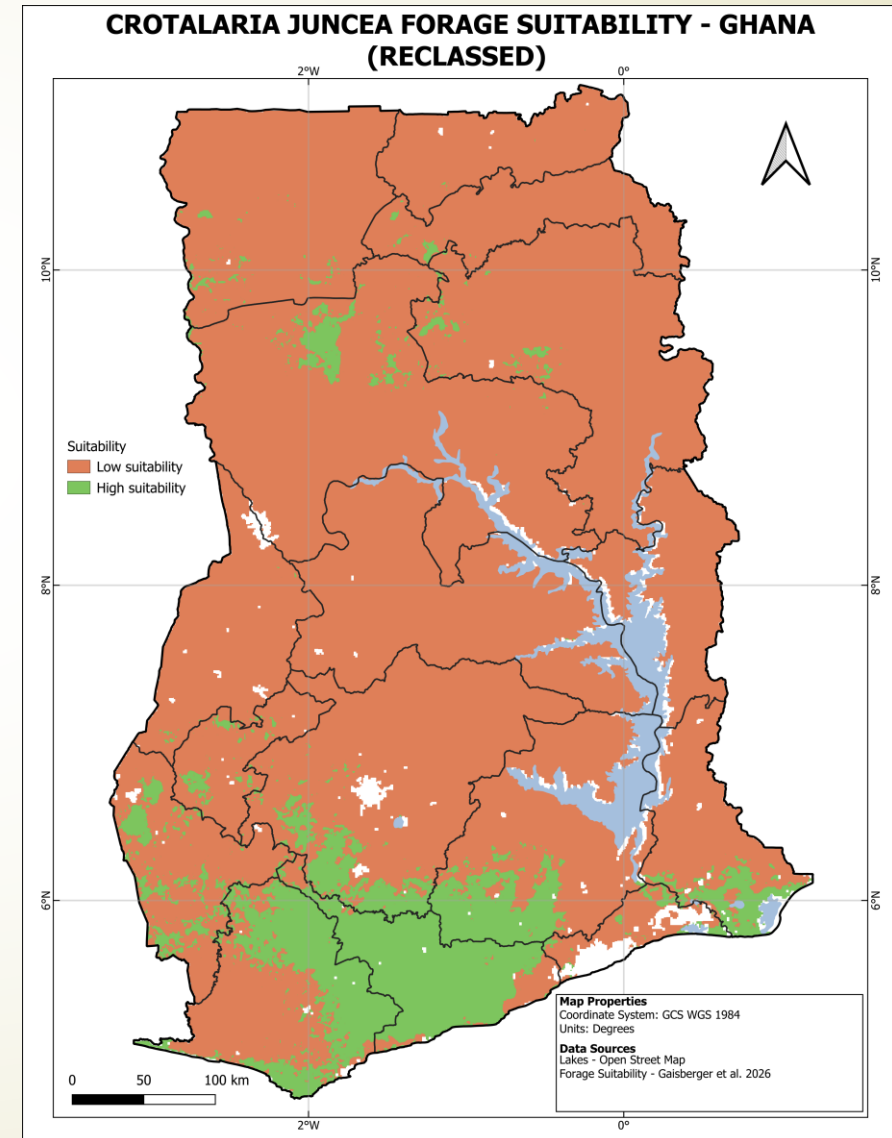
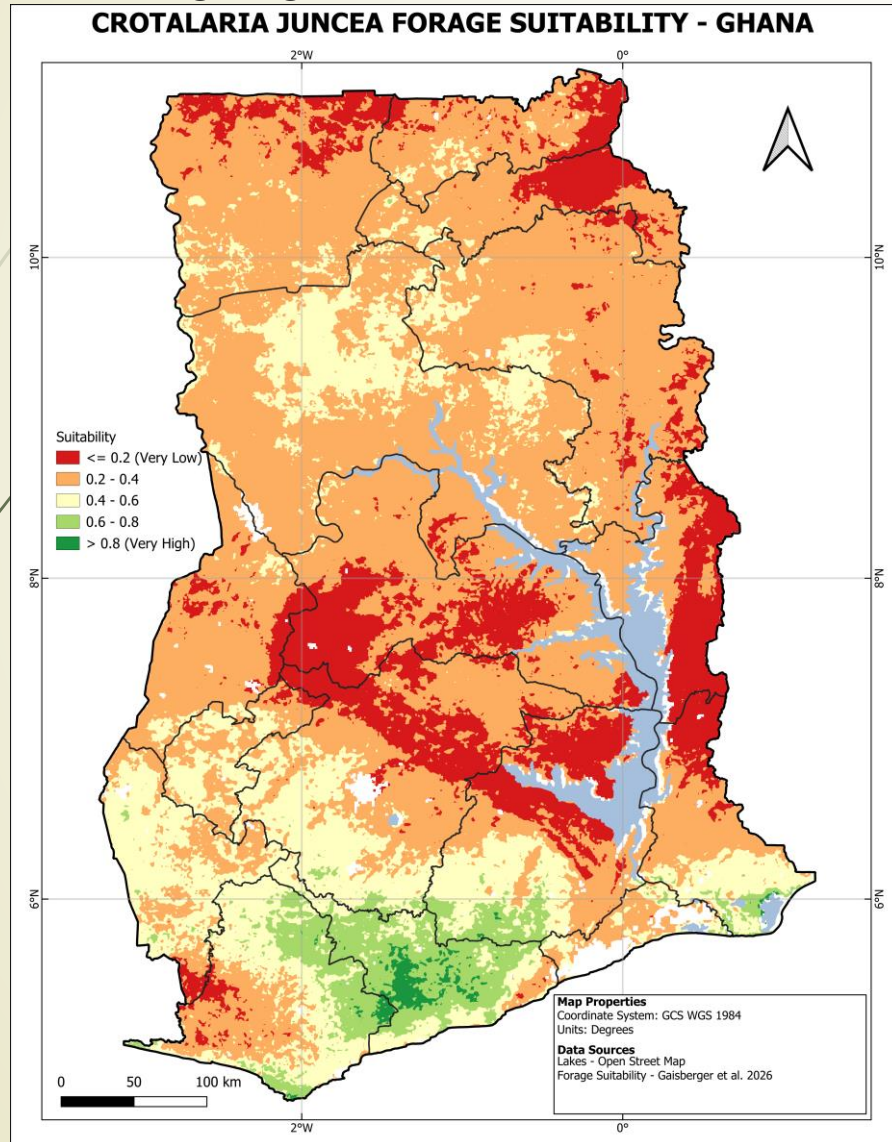
Forages

The forage suitability map help to identify regions suitable for cultivation of a number of shade-tolerant forage species to support livestock production. Here : *Lablab purpureus*. Areas in the optimal class (green) correspond to a suitability value greater than 0.5, indicating regions most suitable for cultivation of the forage species.



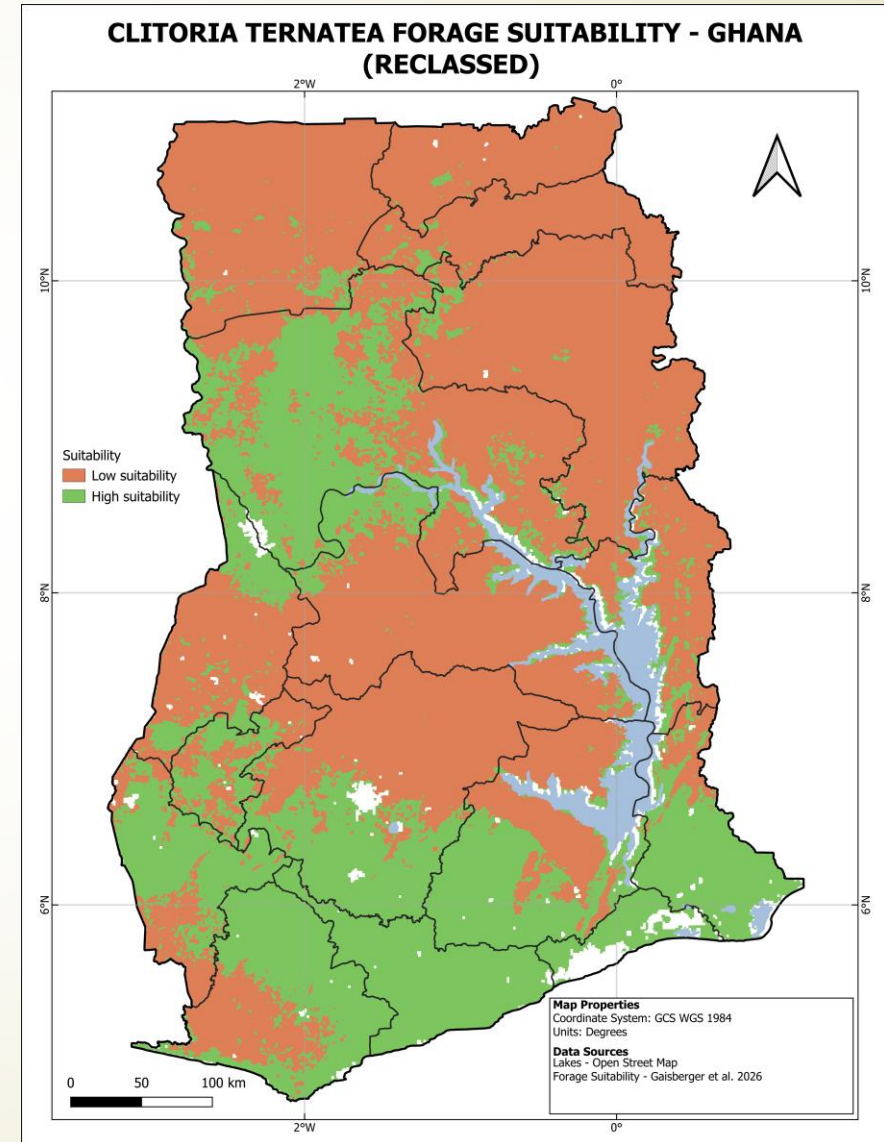
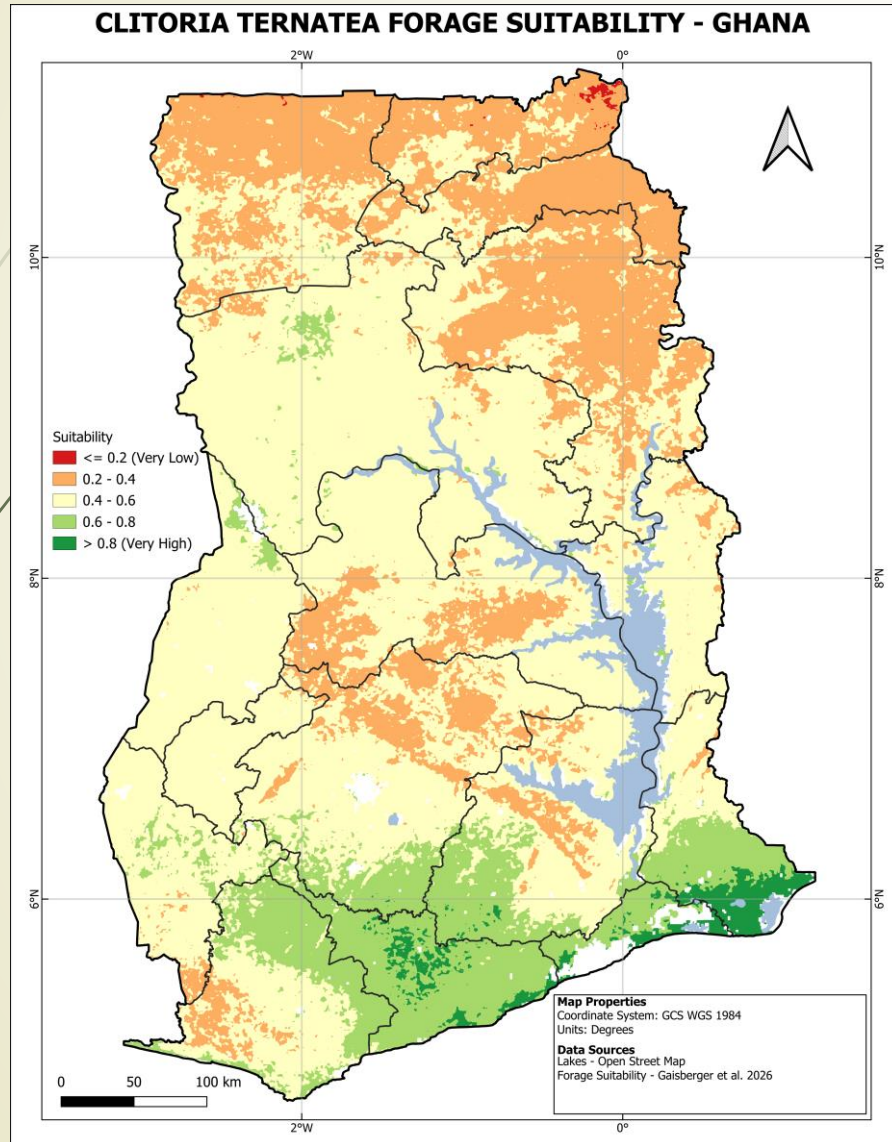
Forages

The forage suitability map help to identify regions suitable for cultivation of a number of shade-tolerant forage species to support livestock production. Here : *Crotalaria juncea*. Areas in the optimal class (green) correspond to a suitability value greater than 0.5, indicating regions most suitable for cultivation of the forage species.



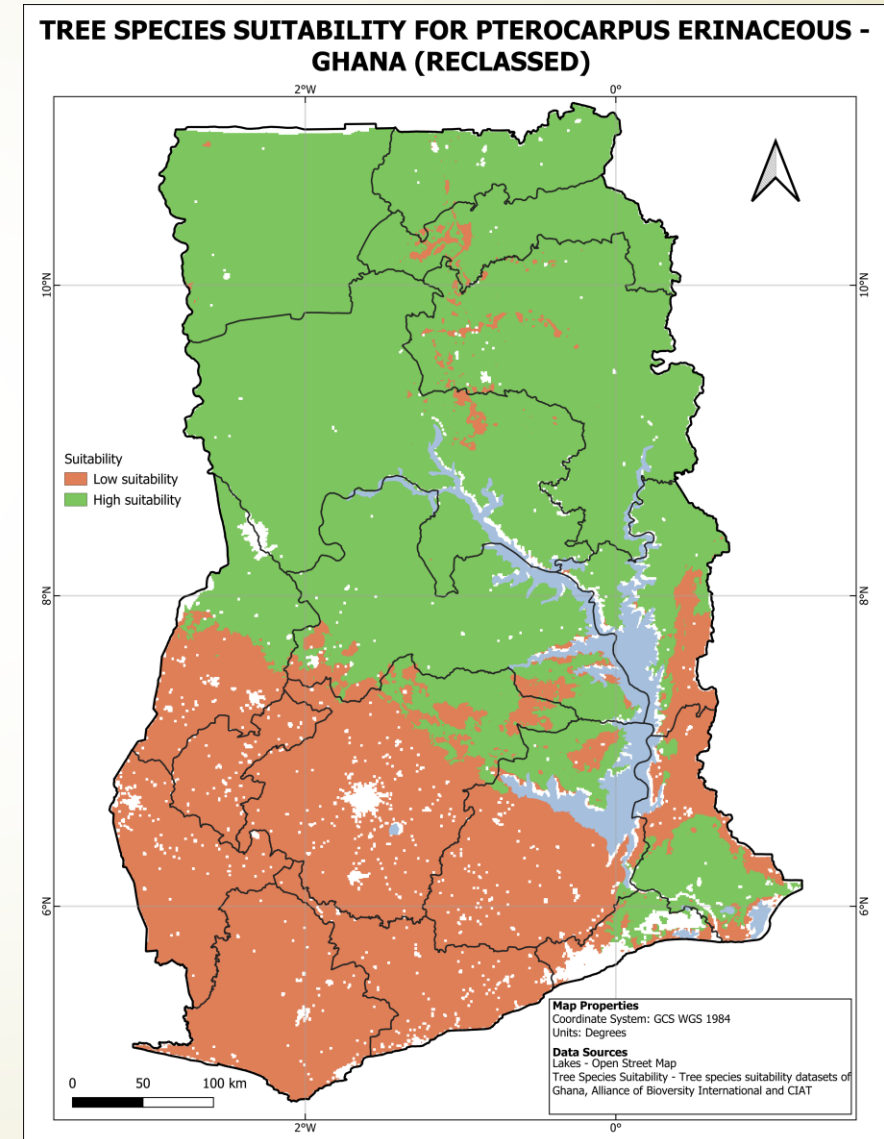
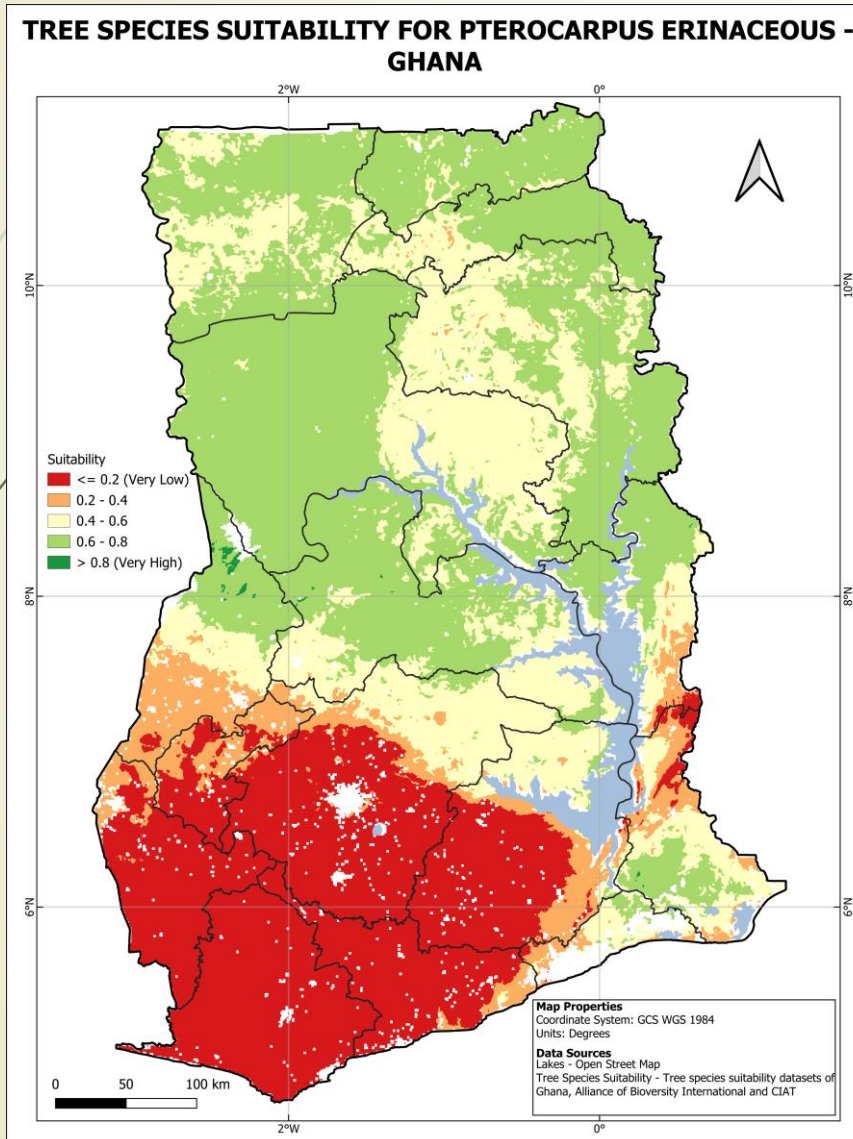
Forages

The forage suitability map help to identify regions suitable for cultivation of a number of shade-tolerant forage species to support livestock production. Here : *Clitoria ternatea*. Areas in the optimal class (green) correspond to a suitability value greater than 0.5, indicating regions most suitable for cultivation of the forage species.



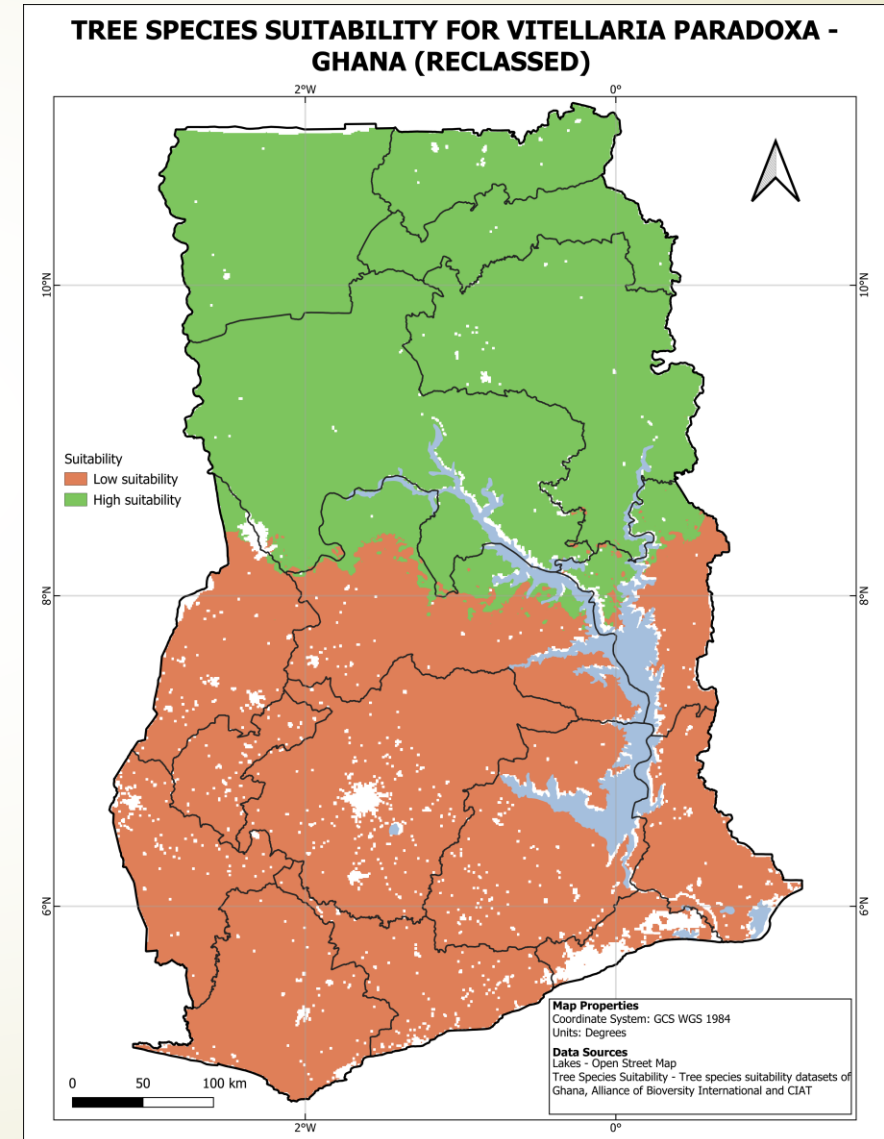
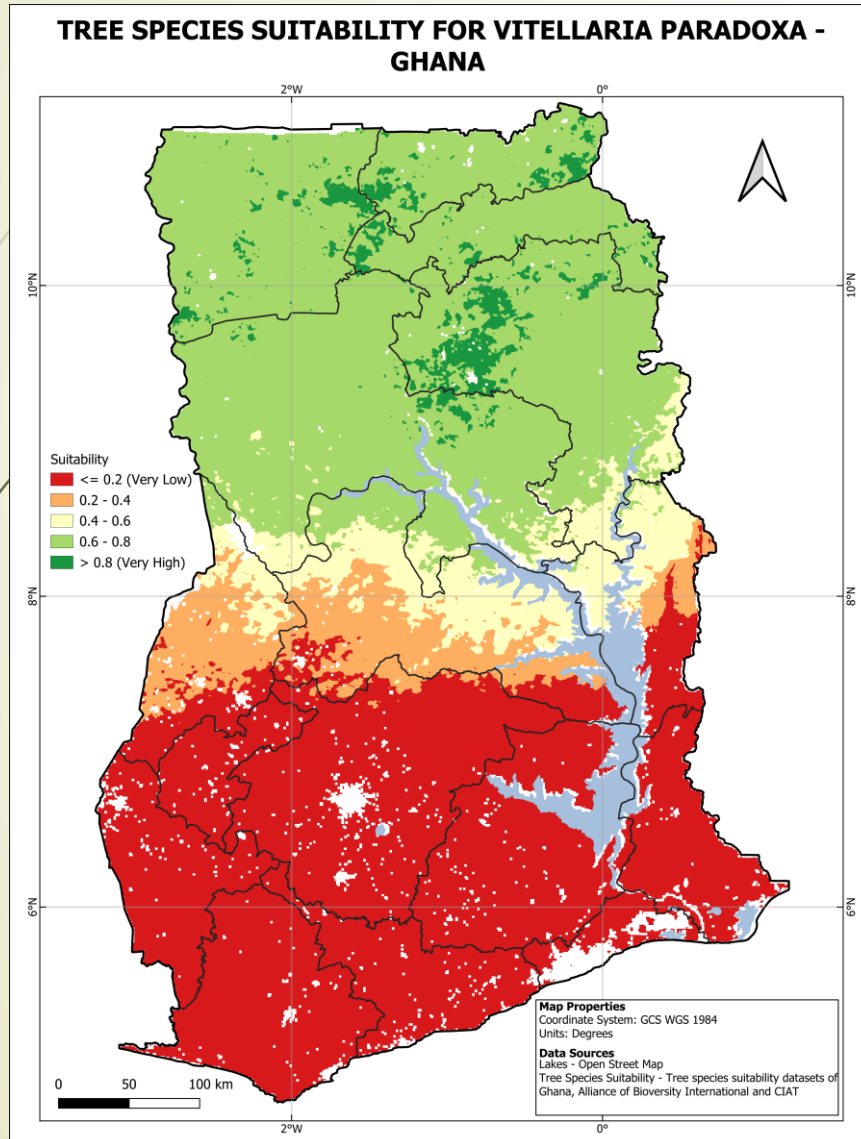
Trees

Tree species suitability map identifies regions suitable for planting tree species commonly reported as fodder for small ruminants in Ghana: *Pterocarpus erinaceus*. High suitability (green) indicates optimal locations corresponding to a suitability value greater than 0.5.



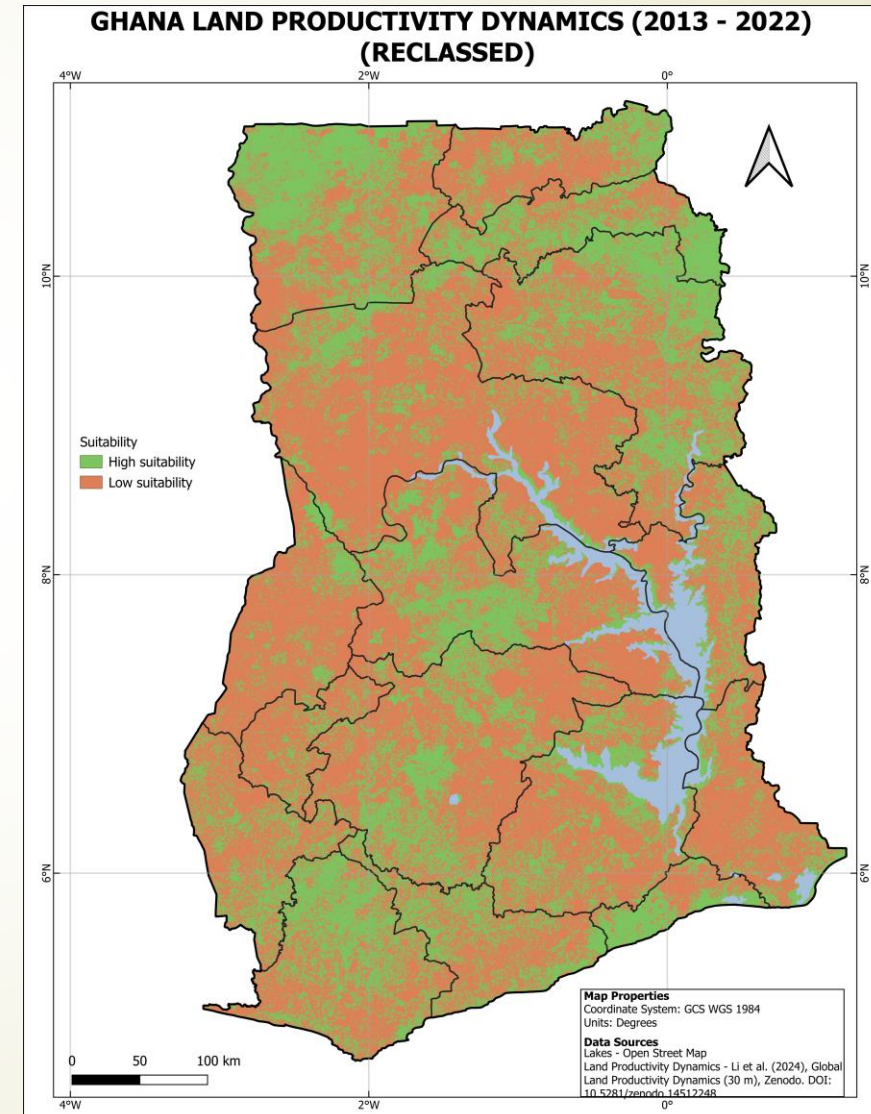
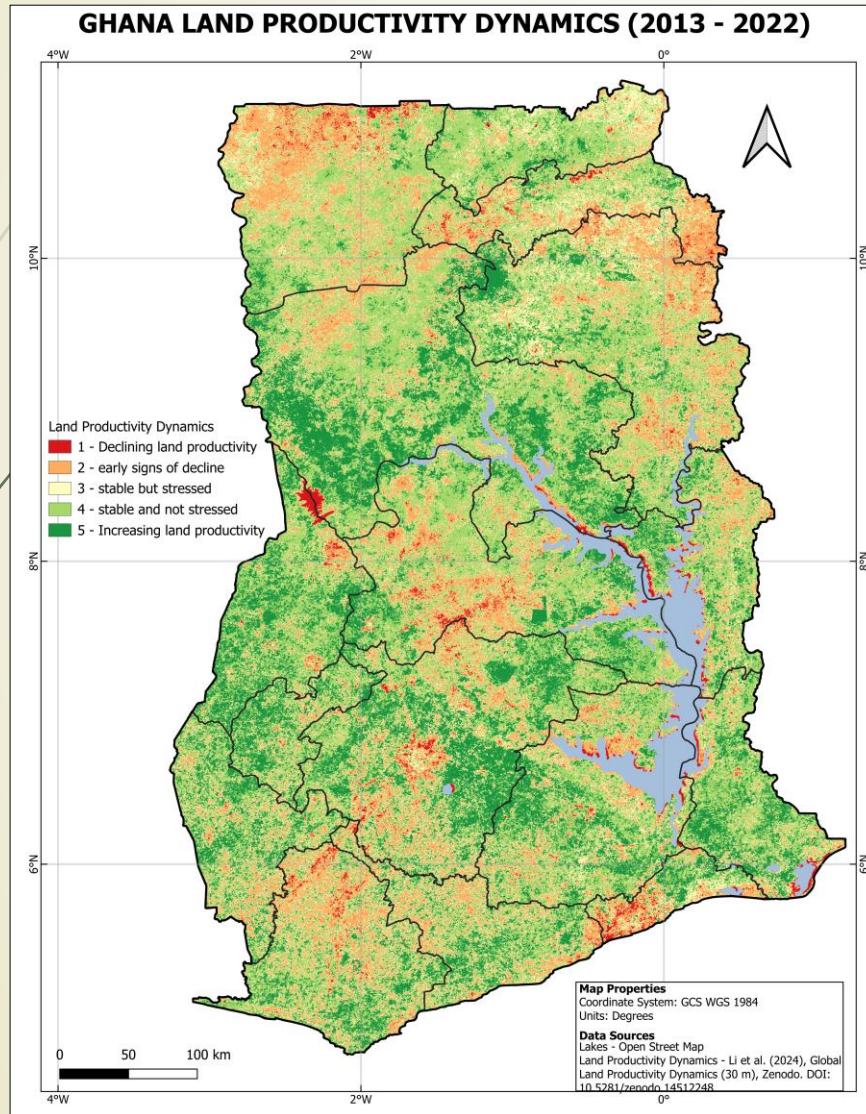
Trees

Tree species suitability map identifies regions suitable for planting tree species commonly reported as fodder for small ruminants in Ghana: *Vitellaria paradoxa*. High suitability (green) indicates optimal locations corresponding to a suitability value greater than 0.5.



Degradation

This map represents land productivity dynamics in Ghana. It highlights areas experiencing land degradation, helping identify regions where silvopastoral interventions could improve productivity and restore degraded lands. High suitability (green) corresponds to areas that are stable but stressed, in early decline, or already declining (values 1–3).

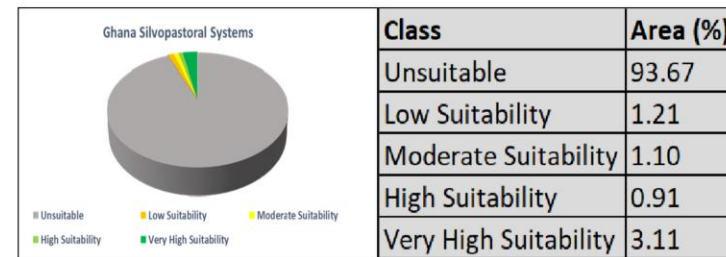
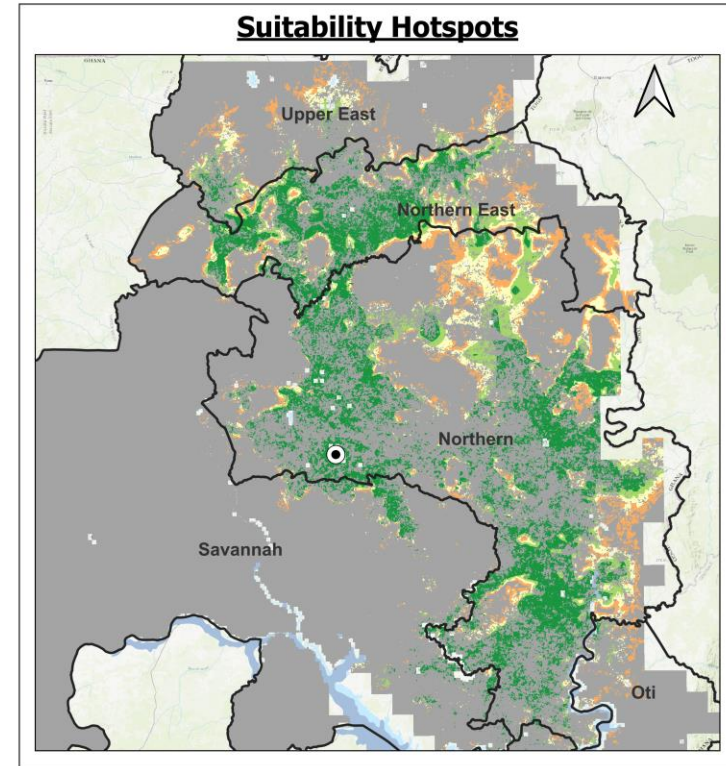
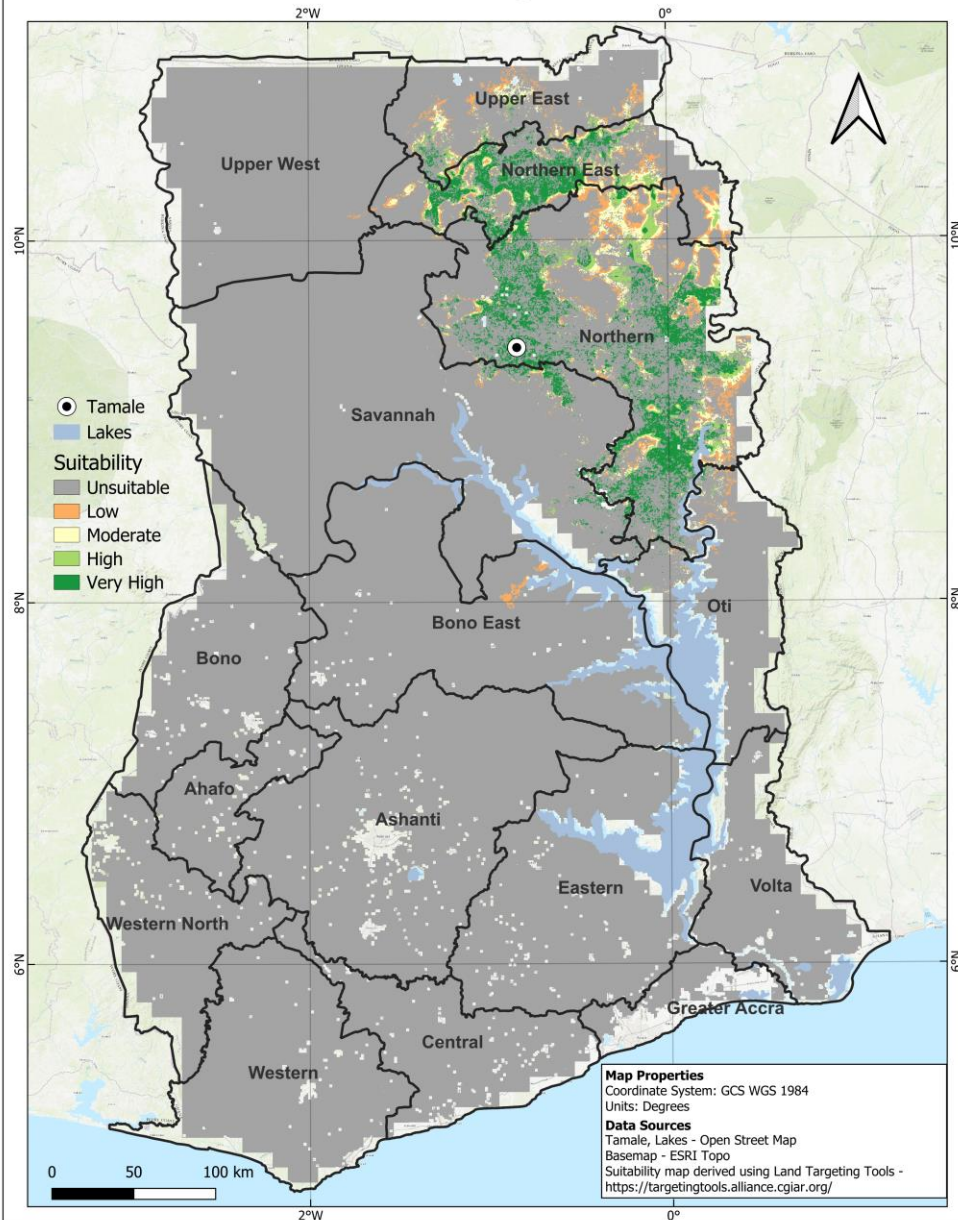


Targeting Thresholds

Factor	Absolute Min	Optimal Min	Optimal Max	Absolute Max
Closeness to Tamale – Travel time to Tamale (minutes) by Car Values range: 0 to 1018 min	0	0	180	240
Family labour – female population density (people/km ²) Values range: 0 – 8000	5	10	8000	8000
Small ruminant density (head/km ²) Values range: 0-850	20	50	850	850
Access to all markets – Travel time to urban areas (minutes) by Car Values range: 0 to 630 min	0	0	180	240
Suitability of shade tolerant forages Values range: 0 to 1 for any of the forages	0.5	0.5	1	1
Suitability of appropriate tree species Values range: 0 to 1 for any of the trees	0.5	0.5	1	1
Degradation layer – Land Productivity dynamics <u>Values</u> 1 - declining land productivity 2 - early signs of decline of land productivity 3 - stable but stressed land productivity 4 - stable and not stressed land productivity 5 - increasing land productivity.	1	1	3	3

Suitability Map

Suitable Areas for Piloting Smallholder Small-ruminant Silvopastoral Systems - Ghana



Targeting Tools



CGIAR
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Statistics – Forage Species

Within the suitable class, ***V. unguiculata*** exhibits a suitability range of 0.69 to 0.91, with a mean of 0.82 to 0.83. In comparison, other forage species have mean suitability values below 0.5, indicating that *V. unguiculata* is the most favourable species across the suitable areas for the silvopastoral system.

Species	Suitability Class	Minimum	Mean	Maximum
<i>Vigna unguiculata</i>	High	0.70	0.82	0.91
	Very High	0.69	0.83	0.91
<i>Urochloa brizanta</i>	High	0.17	0.40	0.67
	Very High	0.13	0.36	0.69
<i>Megathyrsus maximus</i>	High	0.20	0.30	0.54
	Very High	0.19	0.31	0.52
<i>Lablab purpureus</i>	High	0.16	0.27	0.41
	Very High	0.16	0.29	0.45
<i>Crotalaria juncea</i>	High	0.13	0.29	0.60
	Very High	0.13	0.32	0.61
<i>Clitoria ternatea</i>	High	0.22	0.37	0.59
	Very High	0.23	0.39	0.6

Statistics – Tree Species

Within the suitable class, **V. paradoxa** shows a suitability range of 0.46 to 0.85, with a mean of 0.72 to 0.73 while **P. erinaceous** ranges from 0.29 to 0.76 with a mean of 0.60 to 0.62. Both species have suitability values above 0.5, indicating they are favourable across the suitable areas for the silvopastoral system.

Species	Suitability Class	Minimum	Mean	Maximum
<i>Vitellaria paradoxa</i>	High Suitability	0.47	0.73	0.84
	Very High Suitability	0.46	0.72	0.85
<i>Pterocarpus erinaceous</i>	High Suitability	0.31	0.62	0.76
	Very High Suitability	0.29	0.60	0.76



Conclusion

- ▶ This study demonstrates that there are a few areas near Tamale that are well suited and practical for piloting small ruminant silvopastoral systems.
 - ▶ The Land Targeting Tools prove to be useful tools for spatial targeting analyses integrating both bio-physical and socio-economic data.
 - ▶ With vast applications to be considered, the tools support informed decision making for the planning and promotion of silvopastoral systems.
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- ▶ This work was supported by the CGIAR Sustainable Farming (SF) and Multi-Functional Landscapes (MFL) Science Programs. They form a part of CGIAR's new Research Portfolio.
- ▶ SF addresses key challenges in agri-food systems by fostering efficient production of nutritious foods and safeguarding the environment to create fair employment opportunities, while simultaneously tackling climate change, soil degradation, pests, diseases, and desertification.
- ▶ MFL enhances the resilience, productivity, and sustainability of agricultural landscapes by integrating diverse land uses, ecosystem services, and livelihood strategies, thereby supporting evidence-based policies and innovations that balance food production with climate adaptation, biodiversity conservation, and social inclusion.
- ▶ We would like to thank all funders who supported this research through their contributions to the CGIAR Trust Fund: <https://www.cgiar.org/funders/>